DESIGN AND ASSEMBLY OF ORNITHOPTER

Project report submitted in partial fulfilment of the Requirements for the degree of

Bachelor of Technology in Mechanical Engineering

In

Mechanical engineering

SUBMITTED BY

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This is to certify that the project work entitled **"Design and Assembly of ornithopter"** is being submitted by "G.Abhiram (18311A03E1), Sai Krishna.G (18311A03G3)&Syed Sameer(18311A03G6) in partial fulfilment of the requirement for the award of the degree of **Bachelor of Technology** in **Mechanical Engineering,** during the academic year **2021-2022**

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DECLARATION

We here by certify that the project report entitled **DESIGN & ASSEMBLY OF ORNITHOPTER** being submitted by us is worked out in Sreenidhi Institute of Science and Technology under the guidance: Dr.A.Purushotham The results embodied in this project report have not been submitted to any other University or

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ABSTRACT

The ornithopter is a mechanically designed robot that uses a flapping wing mechanism to achieve flight. These designs are inspired by natural aviators such as birds and insects known for their ease of flight and manoeuvrability in the open sky. However, achieving this feat has presented a challenge due to the complexity of understanding its working dynamics. They autonomously vary the flapping rate and adjust the orientation of their tail to sustain steady flight in the atmosphere. Many experiments were conducted to develop the precise mechanism, to help thoroughly understand the dynamics of its machinery, and solving the enigma of its sustained flight in the air. The main objective of the project is to mimic the fluttering actions of a natural bird and understand its flight patterns. Recreating a sizeable model capable of manifesting the same functions as a bird but with other additional features such as surveillance, image capturing. These improvements have wide applications in the military field, research field, and general public applications. This model can be controlled over a long distance using a remote control. Hence, the altitude and the model speed can be controlled by the same. The interest rate on this subject has increased in developing ornithopter based on different aviators. Different methodologies were initiated to comprehend how to maintain long-range coverage, use of solar energy to recharge its power source and to sustain long duration flight and working

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CHAPTER -1

INTRODUCTION

1.1 Introduction to ornithopter

Ornithopter flight, or generally known as bird flight, has only 2 degree-of-freedom (DOF), in which the first one is the main flapping motion and the other one is the slight deviation from the stroke plane (out-of-plane flapping motion). As compared to insect flight, which consists of 3 degree-of-freedom, the third degree, the active wing rotation is replaced by passive wing rotation in ornithopter flight (also known as 'feathering'). Passive rotation is caused by the mass inertial force of the wings during flapping. In turn, it simplifies the kinematics of ornithopter flight, as if the wing rotation works automatically.

As micro air vehicles MAVs decrease in size, the complexity of fabrication and airflow characteristics within the Reynolds number range of operation increases as well. A red-tailed hawk has been taken as an inspiration to develop a small-scale ornithopter with single flapping frequency for simple and dominant flapping-wing motion. The main difficulty of ornithopter design comes from the fact that there is almost no knowledge about what the Most important design parameters are and how they affect each other to flapping-wing flight dynamics. Among other optimization analyses, findings show that a tail-wing adoption could improve ornithopters longitudinal flight stability in the presence of continuous flapping motion of its wings

1.2 Definition of ornithopter

- An ornithopter is an aircraft that flies by flapping its wings.

- Designers seek to imitate the flapping-wing flight of birds, bats, and insects. Though machines may differ in form, they are usually built on the same scale as these flying creatures.

1.3 Working of ornithopter

The working of ornithopter is based on natural flapping wings mechanism of biological birds.

In which the rotational power form the BLDC motor is converted in flapping mechanism by using several

Component such as gears, gears box, and pistons. When motor is set to rotate the power and torque is converted form rotational power into liner motion of the Wings in unidirectional.

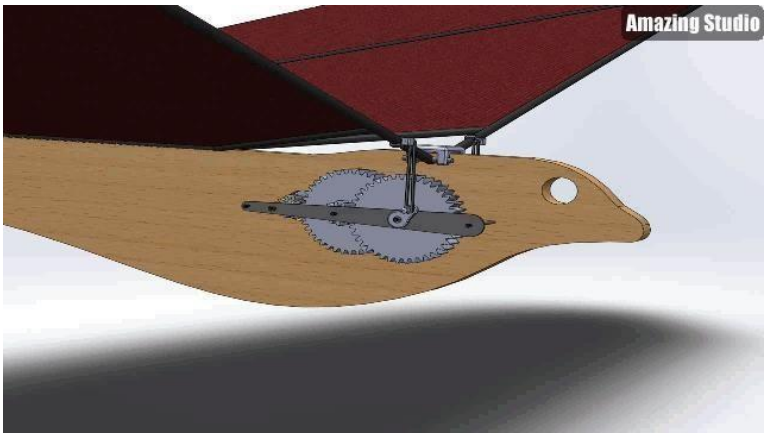


Figure.1 working model of ornithopter

1.4 Application of ornithopter

- Practical applications capitalize on the resemblance to birds or insects. Colorado Parks and Wildlife has used these machines to help save the endangered Gunnison sage grouse. An artificial hawk under the control of an operator causes the grouse to remain on the ground so they can be captured for study.
- Because ornithopters can be made to resemble birds or insects, they could be used for military applications such as aerial reconnaissance without alerting the enemies that they are under surveillance.
- Several ornithopters have been flown with video cameras on board, some of which can hover and manoeuvre in small spaces. In 2011, AeroVironment, Inc. demonstrated a remotely piloted ornithopter resembling a large hummingbird for possible spy missions.
- Researchers hope to eliminate the motors and gears of current designs by more closely imitating animal flight muscles.

1.4 Scope of the project

- Ornithopter is used for military applications such as aerial reconnaissance without alerting the enemies that they are under surveillance.
- Several ornithopters have been flown with video cameras on board, some of which can hover and maneuver in small spaces. These ornithopter mainly uses by spies for information gathering.

1.5 Characteristics of low Reynolds number aerodynamics applied to flapping MAV

At low Reynolds numbers, especially in flapping flight, there are many remarkable results that prove the advantages of unsteady aerodynamics. At the size of insects, lapping wings benefit from unsteady aerodynamics more than steady-state aerodynamics to generate lift, as well as have high manoeuvrability and agility as seen in insects and hummingbirds. Biological flight systems, known as the most efficient flight mechanism, are also superior to engineering flight systems at all small scales for their better power supply, better stability and control system, flying in fluctuating conditions and at low Reynolds numbers.

Small insects have a wing chord Reynolds number between 100 and 1000, use unsteady effects to stay aloft and have corrugated, curved plates for wings. In the range from large insects to small birds, the wing chord Reynolds number lies between 1000 and 15000 and they use conventional air foil circulation and are Sensitive to transition and separation. As the size of MAV decreases with higher wing beat frequency, features of the unsteady flight regime become more critical. In order to fulfil those characteristics, we focused on analysing flight mechanisms and wing structures of insects and adapted those characteristics for our flapping vehicle. As shown in Fig. flight mechanisms and aerodynamic characteristics vary over different Reynolds number regimes. From Pennycuick the relation between flight speed and the mass of a bird can be given by

$$U = 4.77\ m^{1/6} \quad \ldots\ldots\ldots (1)$$

where U is the flight speed in m·s−1 and m is the mass in g. Greenawalt computed from statistical data the correlation between wing flapping frequency f (Hz), vs. wing length l (cm), to be

$fl^{1/6} = 3.45$ ……(2)

While Azuma showed that the correlations between wing flapping frequency f (Hz) and mass, m (g), for large birds and small insects are

$f(\text{large birds}) = 116.3\, m^{-1/6}$ ……(3)

$f(\text{small insects}) = 28.7m^{-1/3}$ ……(4)

From Equation. (1)–(4), relationships between wingtip speed and mass can be derived. These relations are

$\text{Wingtip speed (large birds)} = 11.7m^{-0.065}$ …..(5)

$\text{Wingtip speed (small insets)} = 9.7m^{-0.043}$ ……(6)

1.6 Aerodynamics of flying bird

- As demonstrated by birds, flapping wings offer potential advantages in maneuverability and energy savings compared with fixed-wing aircraft, as well as potentially vertical take-off and landing. It has been suggested that these advantages are greatest at small sizes and low flying speeds, but the development of comprehensive aerodynamic theory for flapping remains an outstanding problem due to the complex non-linear nature of such unsteady separating flows.
- Unlike airplanes and helicopters, the driving airfoils of the ornithopter have a flapping or oscillating motion, instead of rotary. As with helicopters, the wings usually have a combined function of providing both lift and thrust. Theoretically, the flapping wing can be set to zero angle of attack on the upstroke, so it passes easily through the air. Since typically the flapping airfoils produce both lift and thrust, drag-inducing structures are minimized. These two advantages potentially allow a high degree of efficiency.

CHAPTER -2

THEORY ON BIRD FLIGHT

2.1 Bird flight theory

Bird wings are responsible for generating almost all of the aerodynamic forces required for flight, although the body and tail can also generate forces used for flight control and lift enhancement. This arrangement is in contrast to fixed-wing aircraft in which the functions of lift and thrust are decoupled and Integrated, multilevel systems approach to the study of bird flight. Aerodynamic forces are influenced by

wing shape and wing motion. The latter is actively controlled by the neuromuscular system, which is informed by a diverse sensor suite.

Can. J. Zool. Downloaded from www.nrcresearchpress.com by University of British Columbia on 12/04/15
For personal use only. generated by the wings and either propellers or jets, respectively. The ability of birds to orient wing forces into both body weight support and thrust is possible because their wings move with respect to the body. This is facilitated by joints that provide degrees of freedom for wing motion around the shoulder and for active shape change, i.e., morphing, with the arm and hand wing. Aerodynamic force can be decomposed into separate components, and to quantify bird flight performance, we must consider lift, thrust, and drag in a body frame of reference.

2.2 Definition of lift, thrust and drag:

- Lift is defined as the vector component perpendicular to the oncoming air, while thrust and drag are defined as vector components parallel to the oncoming air; thrust points in the direction of motion and drag points opposite. Because the forward body velocity orients the body reference frame in forward horizontal flight, lift supports weight and drag is opposed by thrust. To study the aerodynamic function of the wing, we must consider lift, thrust, and drag vectors in a wing frame of reference. A wing segment is considered in isolation and the force components are defined relative to the local velocity. Total wing velocity during flapping is composed of body velocity, which points forward, and relative wing motion, of which the magnitude and orientation depend on stroke angle, span wise position, and angular velocity.

The wing down stroke is responsible for almost all weight support and thrust generation in the birds that have been studied to date, with the primary exception being hummingbirds. During the down stroke in forward flight, wing velocity points more downward and lift accordingly points more forward, thereby contributing to both weight support and thrust.

- Force components also depend on air density that varies with altitude and incident velocity at the location at which the pressure force effectively acts, the radius of gyration. Lift, drag, and thrust decrease with lower density and increase with greater wing surface area. The coefficients of lift, drag, and thrust are dimensionless numbers that are primarily affected by the shape of the wing and the angle at which it encounters oncoming air, known as the angle of attack. These coefficients are calculated by measuring forces for a given angle of attack, density, wing surface area, and flight velocity typical for a particular species, after which the lift and drag equations can be solved for the coefficients. Because birds fly over a range of speeds that are relatively slow and because their wing-chord lengths are small, the force coefficients depend strongly on the ratio of inertial versus viscous forces in the airflow. Inertia force is proportional to density and viscous forces are proportional to viscosity and the force ratio is estimated with the Reynolds number: **Re Vc·/ ≈ Vc· 68 000 For birds flying near sea level, / ≈ 68 000** and Re ranges between about 5 000 and 15 000 for hummingbirds to about a million for diving falcons. Determining accurate aerodynamic force coefficients is well established for the high Reynolds number of fixed-wing aircraft, but more challenging for the low Reynolds numbers morphing wings of birds. Force coefficient calculations have been successfully applied in hovering and gliding for prepared wings.

2.3 Types of bird wing structures:

1. **Passive soaring wings** have long primary feathers that spread out, creating "slots" that allow the bird to catch vertical columns of hot air called "thermals" and rise higher in the air. Examples of birds with this wing type include eagles, most hawks, and storks. The Bald Eagle is a classic example of a bird with passive soaring wings.
2. **Active soaring wings** are long and narrow, allowing birds to soar, or fly without flapping their wings, for a long time. However, these birds are much more dependent on wind currents than passive soaring birds. Examples of birds with this wing type are albatrosses, gulls, and gannets.

3. **Elliptical wings** are good for short bursts of high speed. They allow fast take offs and tight manoeuvring. As the name suggests, birds with this wing type are incredibly fast, but unlike those with elliptical wings, these birds can maintain their speed for a while. Examples of birds that have this wing type are crows, ravens, blackbirds, sparrows, and thrushes (such as the American Robin).

4. **Hovering wings** are small and quick. For hovering wings, in addition to the wing shape, the bird's nerves and muscles are specially adapted for incredibly fast movement. The Ruby-throated Hummingbird might appear to float in space while sipping nectar and hovering from flower to flower thanks to its quick and mighty wings.

5. **High-speed wings** are long and thin, but not nearly as long as birds with active soaring wings. As the name suggests, birds with this wing type are incredibly fast, but unlike those with elliptical wings, these birds can maintain their speed for a while. Examples of birds that have this wing type are swifts, ducks, falcons, terns, and sandpipers.

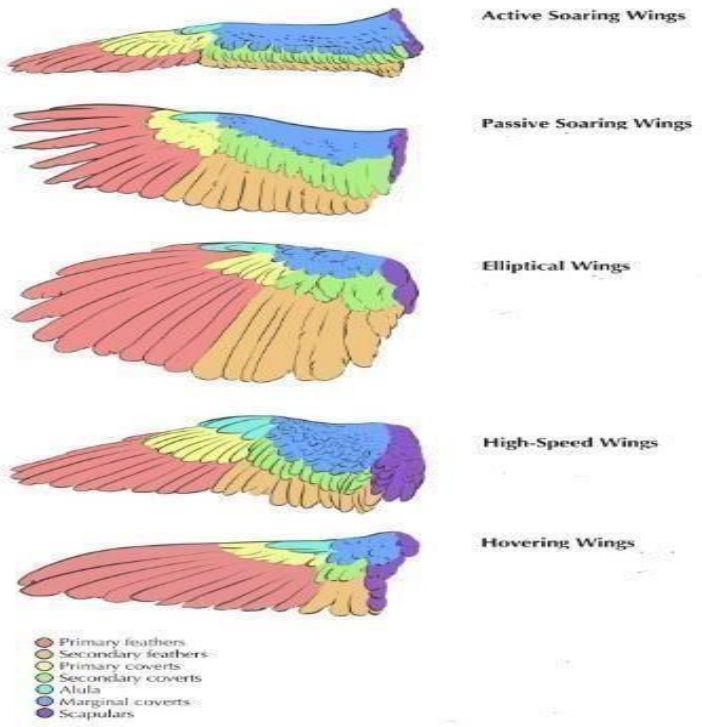


Figure.2 various shapes of bird wings.

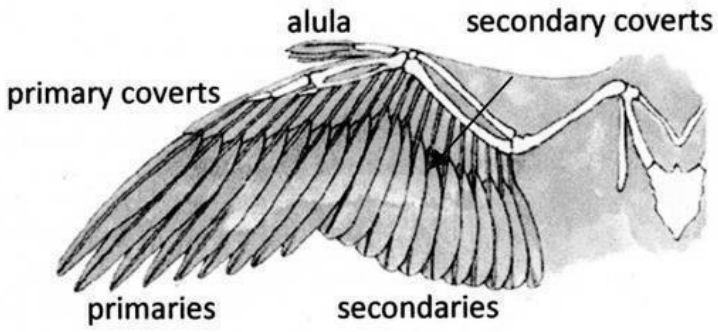


Figure. 3 structure of the bird wing

2.4 Wing aspect ratio

The aspect ratio is the span divided by the mean or average chord.[9] It is a measure of how long and slender the wing appears when seen from above or below.

- **Low aspect ratio**: short and stubby wing. More efficient structurally and higher instantaneous roll rate. They tend to be used by fighter aircraft, such as the Lockheed F-104 Starfighter, and by very high-speed aircraft including the North American X-15.
- **Moderate aspect ratio**: general-purpose wing, very widely used, for example on the Douglas DC-3 transport.
- **High aspect ratio**: long and slender wing. More efficient aerodynamically, having less induced drag. They tend to be used by high-altitude subsonic aircraft such as airliners like the Bombardier Dash 8 and by high-performance sailplanes such as the Glaser-Dirks DG-500.



Low aspect ratio Moderate aspect ratio High aspect ratio

Figure.4 Various aspect ratios of wings

Most variable geometry configurations vary the aspect ratio in some way, either deliberately or as a side effect. For the design of ornithopter wing generally we consider high wing aspect ratio.

2.5 TYPES OF BIRD FLIGHT

Before we take up a discussion of the way in which a bird's wings produce the required forces for flight, it would be useful to take a general look at the main types into which bird flight may be classified. Although practically every species seems to have some specific features, it is easily observed that, broadly speaking, there are two main types:

• flapping flight

• gliding flight.

Flapping flight- "power-on"

This can be regarded as "power-on" flight since the bird continuously expends energy in flapping its wings and generates the thrust and lift forces to propel it and overcome gravity. Careful observation shows that, while the general manner of wing flapping motions may appear similar in all birds, there are many subtle differences between these motions in small and large birds. The motion of the flexible wings is not only in the up-and-down direction but there are also forward and backward components, and parts of the wings twist during the flapping cycle.

Gliding and soaring

Gliding flight occurs when the bird does not flap its wings but uses gravity to provide the means for flight. In still air the path of the bird with wings spread is inclined downwards and the combination of the aerodynamic forces generated by the motion through the air and the force of gravity create a balance of forces allowing steady gliding flight. Soaring takes place when a gliding bird is able to use air currents with vertical velocity components such as thermals, slope currents near mountains, cliffs etc. A particular form of soaring, known as dynamic soaring, can occur only when the wind has a velocity gradient with the speed increasing in the upwards direction.

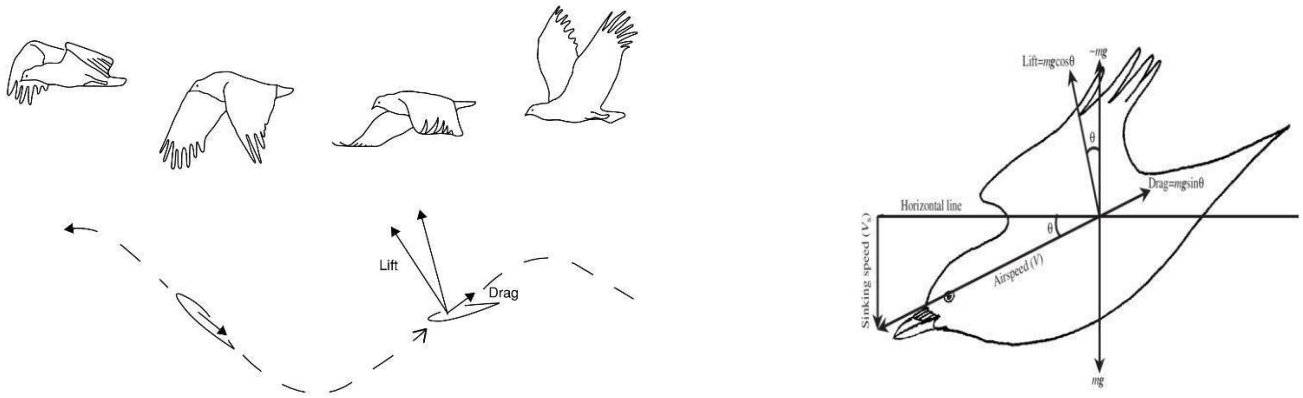


Figure 5. flapping and gliding motion of bird wings.

- When a bird is gliding, it doesn't have to do any work. But it can't stay in the air forever! The wings are held out to the side of the body and do not flap. As the wings move through the air, they are held at a slight angle, which deflects the air gently downward. Pushing the air downward causes a reaction force in the opposite direction. You will notice a reaction force, any time you push against anything! The reaction force is called lift. Lift is a force that acts roughly perpendicular to the wing surface and keeps the bird from falling.
- Soaring flight can only occur at special places and times. For example, warm air heated by the sun can rise up from the hot ground and into the sky. This rising air current is called a thermal. Thermals often rise up along the slope of a hill, but they can also form over flat ground. As the air rises, it also expands and cools. (This result from the lower pressure at altitude.) Eventually, the water vapour in the thermal may become cold enough to condense, forming the tiny droplets of liquid water that make up a cloud. When you have a sunny day with puffy clouds, it's probably a good day for thermals, and you may see hawks or vultures soaring overhead.

CHAPTER-3

THRORY OF AEROFOIL

3.1 Aero foil theory

An aerofoil-shaped body moving through a fluid produces an aerodynamic force. The component of this force perpendicular to the direction of motion is called lift. The component parallel to the direction of motion is called drag. Subsonic flight aerofoils have a characteristic shape with a rounded leading edge, followed by a sharp trailing edge, often with a symmetric curvature of upper and lower surfaces. Foils of similar function designed with water as the working fluid are called hydrofoils.

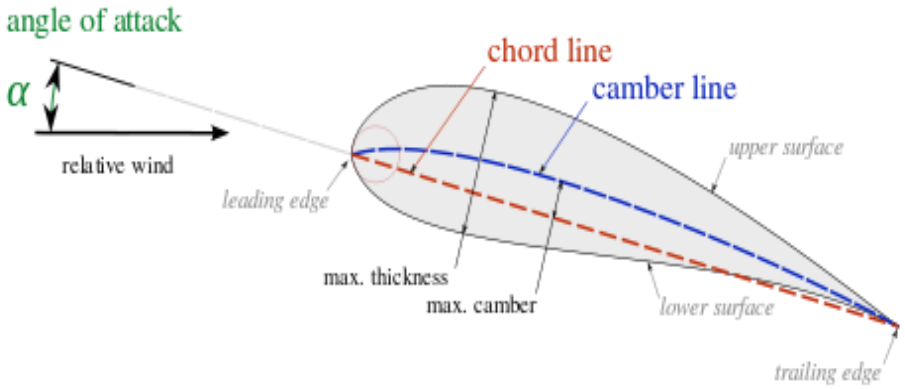


Figure .6 nomenclature of aerofoil

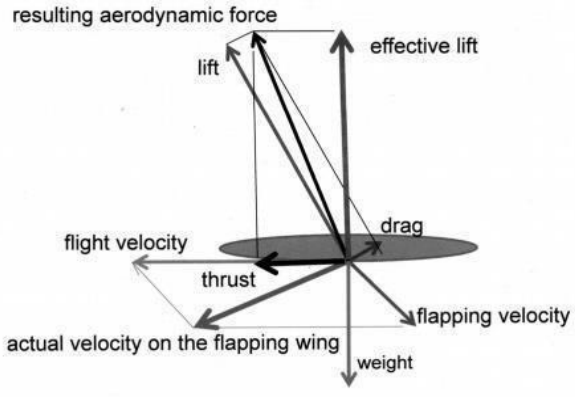


Figure 7 forces acting on aerofoil

3.2 How do birds actually generate lift?

To fly, wings impart downward momentum to the surrounding air and obtain lift by reaction. This is Newton´s 3rd Law, which says that action and reaction are equal and opposite to each other. The lift force, which counterbalances the weight, is obtained from Newton´s 2nd Law.

To obtain the downward momentum, we have to know the downward component of the velocity deflected by the wing and the mass flow which is proportional to the air density, the wing area and the downward component of the velocity. This velocity component depends on the flight speed and on the angle of attack. The mass flow due to the deflected wing depends on the wing area, wing curvature, and in birds also on various lift-increasing-manipulations with the feathers. However, in principle, lift depends primarily on the angle of attack! With the only exception of a hummingbird, birds generate lift and thrust by flapping the wings. This is a complex unsteady and three dimensional motion of the wings, changing at every instant with the new position of the wings. The aerodynamic analysis of bird flight was usually based on the quasi-steady assumption, according to which all instantaneous forces on a flapping wing in unsteady motion are assumed to be those corresponding to steady motion at the same instantaneous velocity and attitude. This assumption may be misleading (as is definitely the case when dealing with insect flight) and it is recommended to consider even the aerodynamics of the bird flight as unsteady. Of course, wings not only generate lift and thrust, they cause drag as well. Optimally designed wings have to bring the air from upstream to downstream with as little loss of momentum, i.e. loss of drag, as possible and at the highest lift as possible. During the down stroke phase of high aspect ratio wings typically two counter rotating vortices are generated – on the leading edge and on the trailing edge, both being shed with the stream. This vortex couple drives the air downward, generating thus lift in the opposite direction. On low aspect ratio wings the tip vortices on either wings dominate, forming again a couple to generate lift.

CHAPTER-4

Flapping Wing Mechanism

The flapping wing mechanism function is to convert the motor's rotary motion into flapping motion. It is the most important component of the MAV thus much research was done to assess the many different designs available. Generally, the mechanism design is about the same to each other with only slight modifications.

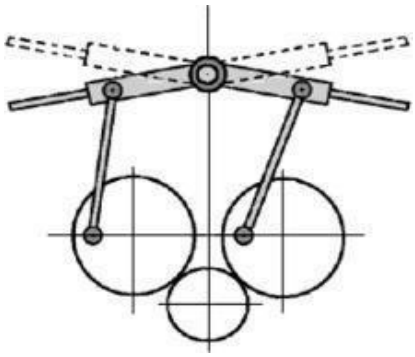


Figure. 8 Basic mechanism of flapping wings

4.1 Staggered Crank Design

The staggered crank design is the most basic of the flapping wing design.

The connector rods are staggered in a measured distance and angle to ensure that the left and right wing are flapping symmetrically. This design is favoured by a hobbyist who wants to attempt to make their own Ornithopter using household items. Modifications have to be made so that the motor can be used instead of a rubber band as its power source.



Figure.9 Staggered crank.

4.2 Single Gear Crank Design

The single gear crank design in taken from University of California Biomimetic Milli systems looks simply however, it is more complicated than it seems. Figure shows the wings at the same level. The centre points where the connector rod and the wing hinges are connected to each other has to expand and contract as the mechanism flaps.



Figure .10 Single gear crank.

4.3 Dual Gear Crank Design

It features two gears that controls each wing hinges separately. There are different variation to the drivetrain design. The one shown in Figure, uses the pinion wheel to drive both the secondary gears. The secondary gears will rotate in the same direction with each other. In the other design, the pinion gear rotates the secondary gear and this secondary gear rotates another secondary gear. The secondary gears would rotate counter clockwise to each other. This design is much simpler to implement and reduce the wing symmetry misalignment.

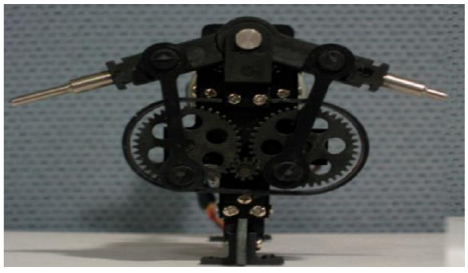


Figure.11 Dual gear crank.

4.4 Transverse Shaft

The transverse shaft design shown in Figure is the other variation of flapping mechanism from which allows for the most symmetrical flap, however, it is the heaviest and the most complicated design. The rotating gears and the flapping wings are not in the same plane thus the connector rod has to be able to rotate.

The connector rod has a ball bearing inside and this adds weight to just the component itself. The number of gears used in this design is more than any other design. The transverse shaft design is usually used for a bigger MAV design where weight could be overcome by large wings.

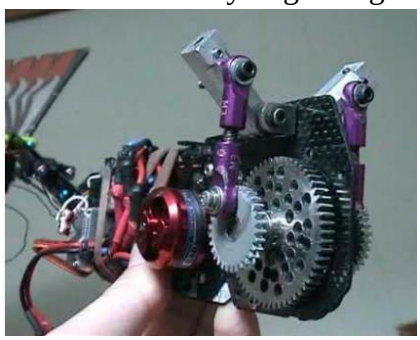


Figure.12 Transverse Shaft.

CHAPTER- 5

SELECTION OF ELECTRONICS

The selection of electronics based on the required power to be produced by the ornithopter that to generate lift, and to control the motion and direction of the flight of the ornithopter.

5.1 Selection of motor

- To produce a specific flapping rate of the wings a brushless direct current (BLDC) motor is attached with designed gears which transfers the rotational power into reciprocation motion to the piston.
- A brushless DC motor (also known as a BLDC motor or BL motor) is an electronically commuted DC motor which does not have brushes. The controller provides pulses of current to the motor windings which control the speed and torque of the synchronous motor.



Figure.13 BLDC motor

Specifications of the motor

- Power : 1800 kv
- Rated voltage : 7.4 volts
- Current : 0.8 A
- Speed :3500 rpm (no load)
- Motor size : 27.7×26.3 mm.

Circuit diagram of BLDC motor

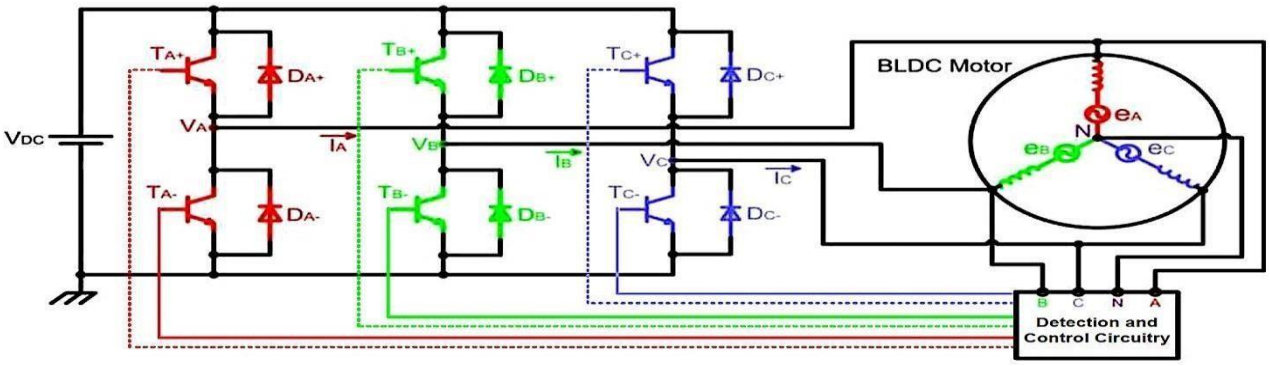


Figure.14 Circuit diagram of BLDC motor

5.2 Selection of speed controller

- An ESC or an Electronic Speed Controller controls the brushless motor movement or speed activating the appropriate MOSFETs to create the rotating magnetic field so that the motor rotates.



Figure.15 electronic speed controller (ESC)

- The higher the frequency or the quicker the ESC goes through the 6 intervals, the higher the speed of the motor will be.
- An electronic speed control or ESC is an electronic circuit that controls and regulates the speed of a electric motor. It may also provide reversing of the motor and dynamic braking. Miniature electronic speed controls are used in electrically powered radio-controlled models. Full-size electric vehicles also have systems to control the speed of their drive motors.

Circuit diagram of electronic speed controller

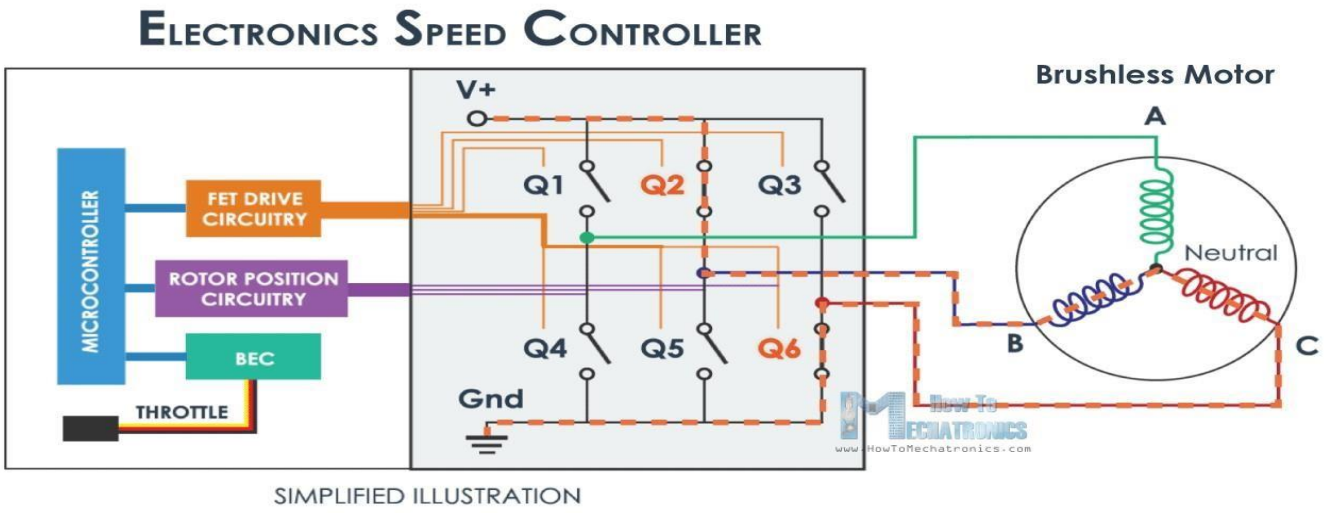


Figure. 16 circuit diagram of electronic speed controller

5.3 **Selection of transmitter and receiver**

- Radio-controlled or remote-controlled toys, popularly called RC toys, are self-powered and can be controlled from a distance using a remote that works with radio waves.



Figure . 17 Remote or radio controller and receiver

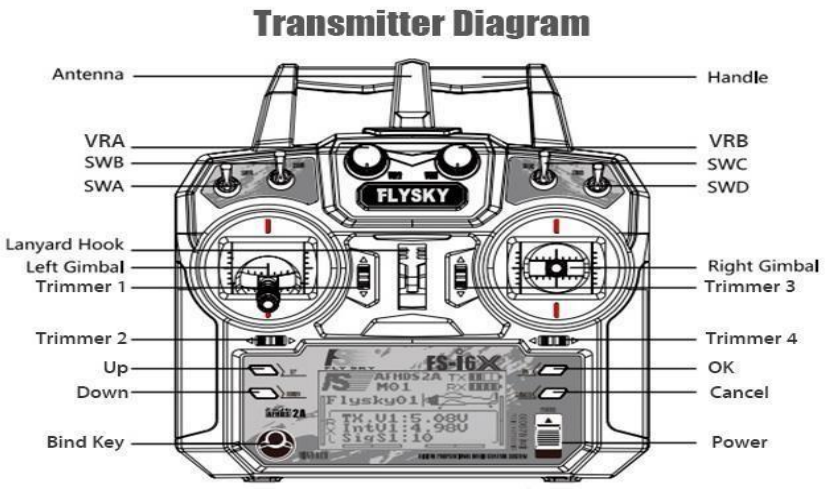


Figure. 18 Transmitter diagram

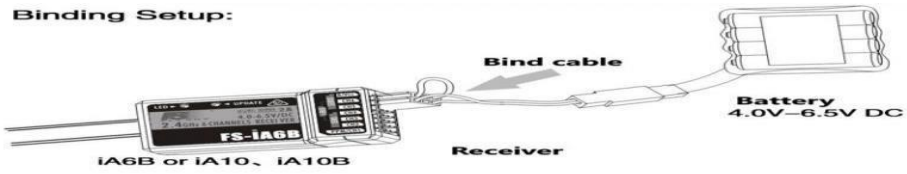


Figure. 19 Binding of transmitter and receiver

- When we push the control, the transmitter sends a specific number of electrical pulses corresponding to that action through the air. The transmitter has its own power source, usually in the form of a 7.5-volt battery. Without the battery, the transmitter will not be able to send the radio waves to the receiver.
- Once the RC toy receives the radio waves, the motors kick into life to cause a specific action to occur. The power source sends power to all working parts, including the motor. The transmitter enables control through radio waves and the receiver activates the motors.
- When we press a button on the transmitter to make the RC toy go forward or backward, a pair of electrical contacts touch. Receiver identifies signals, sends it to circuit.

Specifications and Features:

Specification for FS-i6X RC Transmitter:

1. Item: FS-i6X RC Transmitter
2. Tx Channel: 6
3. Model Type: Fixed-Wing/Glider/Helicopter
4. RF Range: 2.408-2.475GHz
5. RF Power : < 20dBm
6. RF Channel: 135
7. Bandwidth : 500KHz
8. 2.4GHz System : AFHDS 2A / AFDHS
9. Modulation Type: GFSK
10. Stick Resolution: 4096
11. Low Voltage Warning: < 4.2V
12. DSC port: PS/2 Port PPM
13. Chargeable: No
14. Antenna Length: 26mm (Dual Antenna)
15. Weight: 392g
16. Power : 6V DC 1.5A
17. Display: STNTransflective Display, LCD 128×64 Lattice, VA 73x39mm, LCD with white backlight

Specifications for FS-iA10B RC Receiver:

1. Rx Channels: 10
2. Frequency range :2.4-2 .48 GHz

3. Transmitting power: not more than 20dBm
4. 2.4G modes: automatic frequency second generation digital system
5. Antenna length: 26 mm * 2 (dual antenna)
6. Input power :4.0-6 .5 V DC
7. Dimensions: 47 x 33.1 x 14.7 mm
8. Colour: Black
9. Data acquisition interface: Yes
10. Model Type : Airplane / Glider / Helicopter
11. Compatible Transmitter: Compatible with FS-i4, FS-i6, FS-i10, FS-GT2E, FS-GT2G

5.4 Selection of battery

- Selection of battery is based on the required power that consumed by the all electronic parts.

  And also considering the weight of the battery that to be used, if the weight of the battery is Heavy if effects the total weight of the ornithopter.

- For the design of the ornithopter we select battery with following specifications
  - Current : 900 mah
  - No.of cells : 2s
  - Voltage :7.4 volts



Figure.20 2s cell battery

5.4 Selection of Servo motor

- A servo motor is an electrical device which can push or rotate an object with great precision. If you want to rotate and object at some specific angles or distance, then you use servo motor. It is just made up of simple motor which run through servo mechanism.



Figure .21 servo motor

- Servo motor is used to change the direction of the ornithopter which is connected to the tail Of the ornithopter.
- Two servo motors are fitted to the right and left sides of the V-shaped tail so that specific Signal from the remote control is sent to the servo motor to change the direction of the light of ornithopter.

These electronic parts control the flight and mobilise the ornithopter in specific speed and direction of the Flight for the ornithopter

Specification of 2 *servo motor

Modulation:	Analog
Torque:	4.8V: 25.00 oz-in (1.80 kg-cm)
Speed:	4.8V: 0.12 sec/60°
Weight:	0.32 oz (9.0 g)
Dimensions:	Length:0.91 in (23.0 mm) Width:0.48 in (12.2 mm) Height:1.14 in (29.0 mm)
Motor Type:	3-pole

CHAPTER-6

DESIGN AND FABRICATION OF ORNITHOPTER

Design of ornithopter is distinguished into different part as of following:

1) Design of wings
2) Design of gears
3) Design of gear box
4) Design of fuselage
5) Design of tail
6) Design of frame

5.5 DESIGN OF WINGS

The wing is the member that generates lift. The wing is designed by bio mimicry of falcon wing.

A bird wing is divided into three sections.

Primary section

This is the section of wing which has a pure membrane structure and with largest area. The maximum amount of lift and thrust are produced by this section with a vortex formation.

Secondary section

This section is a combination of aerofoil and membrane. The thick aerofoil gradually turns into membrane moving away from the ornithopter body.

Tertiary section

This is the section near the ornithopter body. It is an aerofoil section with high camber and small area. This section helps the bird with a small amount of lift while gliding and up stroke.

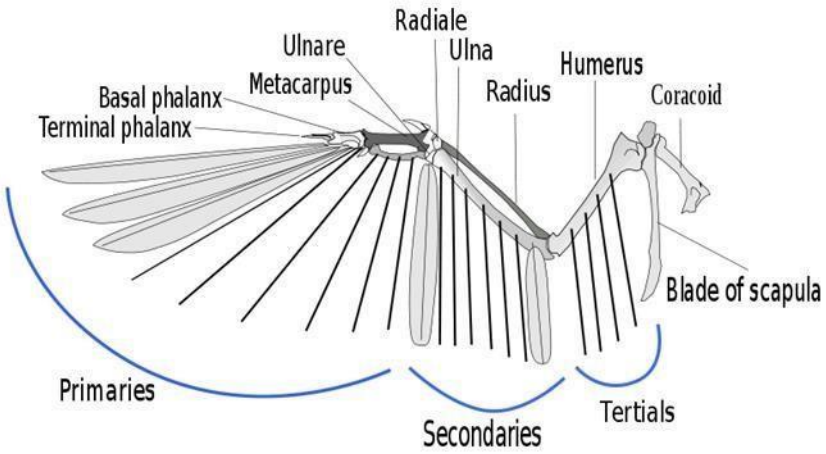


Figure.22 Structural represntation of bird wing

Calculaton of wing span

The wing span of the ornithopter is selected based on thr biological bird references. The references are Made using **RED HAWK** bird

- The wing width is calculated by taking 1/8 part of the fuselage length – total length of the fuselage

  Fuselage length = 400 mm

  W= F-1/8 * fuselage length = 1/8 * 400

  W = 400-50 = 350 mm

  W= 35 cm

- The wing width is also know as chord length.

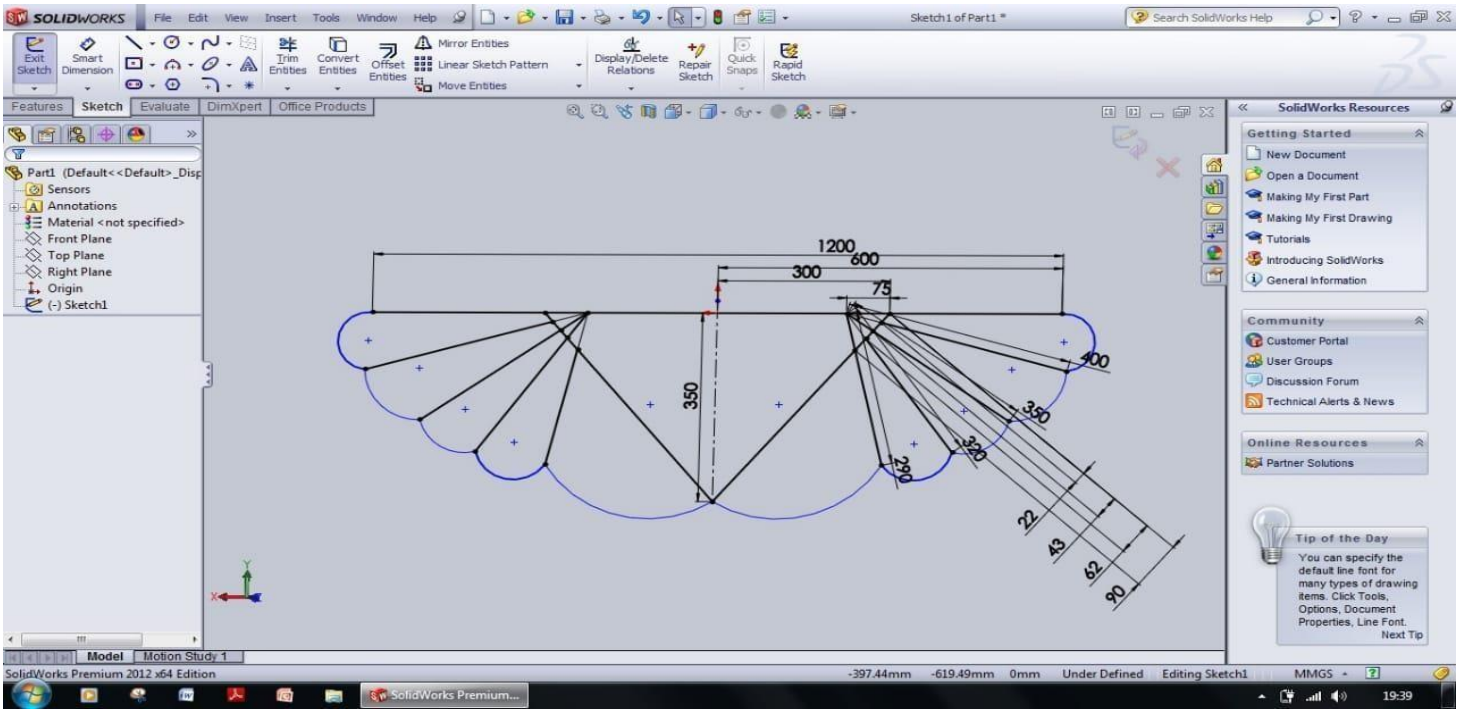


Figure.23 design of wings using solid works software

Fabrication of wings

- In the design and construction of any ornithopter, one of the most crucial components governing overall performance is the wings. Especially in a flapping-wing air vehicle, the wings are vital to flight performance aspects including endurance, speed, manoeuvrability, and many other useful behaviours. An effective ornithopter must have wings capable of generating both thrust, the force that propels the craft forward, and lift, the force, perpendicular to the direction of flight, that keeps the ornithopter airborne. These forces must be strong enough to counter the effects of drag and the weight of the ornithopter.

- The wings produce lift and thrust primarily due to their ability to change shape during the flapping motion. As the wing is accelerated, aerodynamic loading causes the wing to deform. The result is a fairly large camber change of the wing that results in the normally flat plate shape changing to an air foil shape. The aerodynamic loading also produces large angles of attack that create thrust. When these two effects are combined, the air foil wings are placed in a moving airstream, thus creating a flight sustaining lift force.

- Ornithopters with thinner wings have a limited angle of attack but provide optimum minimum-drag performance
- for a single lift coefficient. The Hawk has the closest characteristic as our ornithopter prototype, with the weight of 181.3 g and therefore the dimension of the constructed wing.



Figure.24 Fabricated wings of ornithopter

Materials used for construction

1) Carbon fiber rods for primary spar : 3mm carbon fiber rods
2) Carbon fiber rods for secondary spar : 2mm cabon fiber rods
3) Carbon fiber rods for ribs spar : 1.5 mm cabon fiber rods
4) Wings aerofoil membrane : Epp (expanded poly propylene)
5) Length of the wing span = 1200 mm
6) Chord length = 350 mm

5.6 DESIGN OF GEARS

A **gear** or **cog** is a rotating machine part having cut teeth or, in the case of a cogwheel, inserted teeth (called cogs), which mesh with another toothed part to transmit torque. Geared devices can change the speed, torque, and direction of a power source. Gears almost always produce a change in torque, creating a mechanical advantage, through their gear ratio, and thus may be considered a simple machine.

The teeth on the two meshing gears all have the same shape. Two or more meshing gears, working in a sequence, are called a gear train or a *transmission*. A gear can mesh with a linear toothed part, called a rack, producing translation instead of rotation.

When two gears mesh, if one gear is bigger than the other, a mechanical advantage is produced, with the rotational speeds, and the torques, of the two gears differing in proportion to their diameters.

Calculation of gear reduction ratio for obtaining flapping rate

For the generate of lift by the wing, the flapping rate of the wings should have between ‘6-8’ for the Selected design of the ornithopter.

Where the minimum flapping frequency per second is ‘6’ and the maximum frequency should be ‘8’.

By selecting various gears we can obtain such flapping frequency for the wings from the motor power and Rotational speed of the motor.

Gear reduction is made to calculate the flapping wing rate for required flight.

A. Main gear: 50 teeth
C. Secondary gear: 90 teeth
B. Pinion gear: 10 teeth
D. Driver gear : 12 teeth

Appling gear reduction ratios:

A : B : C : D

(50÷10) × (90÷12) = 1 : 37.5 = 1/37.5

Rotational speed calculation:

7.4 v * 1870 rpm/v = 13,690 rpm (from motor specifications) **Rotation speed per second :**

13,690/60 = 228 (rotation per sec)

Calculation of flapping rate of wings:

Rotational speed of gears * gear ratio

226*1/37.5 = 6.09 (flaps per second)

By the above calculaton the flapping frequency per second is obtained has '**6.08**'

And the required frequency should have between '**6 – 8**'. So the design of gears can obtain by calcutating

The ration reduction rate of the selected gears.

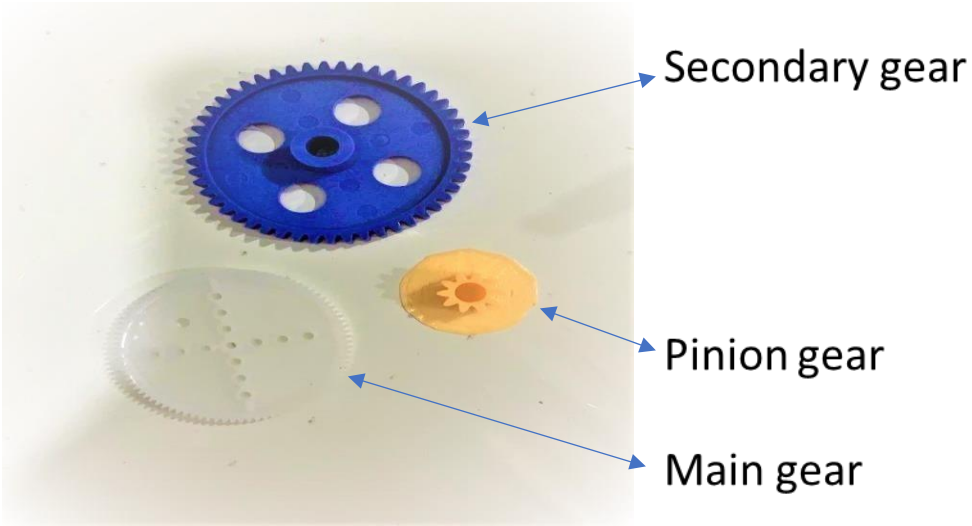


Figure.25 Fabricated gears for the gear box

5.7 DESIGN OF GEAR BOX

- Gear box is a mechanical device where the gears are used for transmitting power from one part of a machine to another.
- The most critical part of the ornithopter is the drive mechanism the converts the electric power from the battery to the flapping motion of the wings. This system is the most complex to design and fabricate because it must withstand very large forces which reverse direction several times a second while at the same time being extremely light and durable.

- The drive system can be further broken down into four sections, the electric motor, a gear reduction stage, a linkage to convert the high torque rotation into a reciprocating motion, and the connection to the wing spars. While a highly integrated design is needed in order to maximize the power to weight ratio, the analysis of these parts breaks down well.
  1) To increase speed and decrease torque
  2) To decrease speed and increase torque
- In the design of ornithopter gear box we need to decrease the speed and increase so the power form the Motor i.e, rotational speed is converted in flapping wing mechanism.
- In ornithopter the gear box is located at front side of the fuselage where gears are mounted in respective Position.
- A duel position made of carbon fibre is attached on both sides of the driven gear which transfer the Rotation motion of the gears in to oscillation motion.
- This oscillation motion is acquired from single crank and sider mechanism of the connecting piston which is connected one end to driven gear and other end to wings main rids using stoppers.

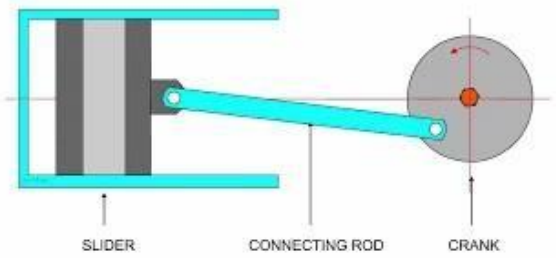


Figure.26 Single slider mechanism of piston connection rod

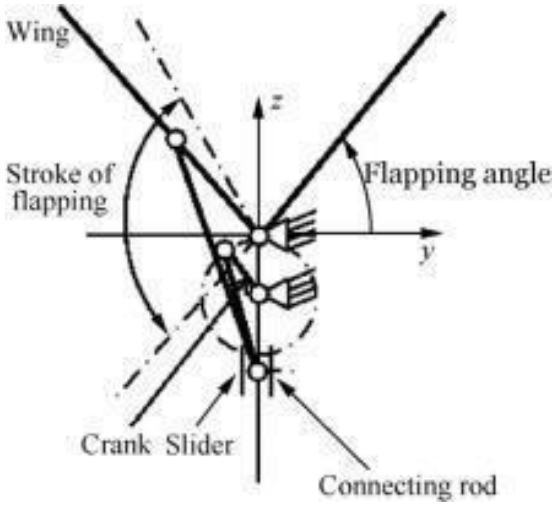


Figure.27 crank and slider mechanism of ornithopter wings

- This gear box is then connected this a BLDC motor where the shaft of the motor is linked with a pinion Gear which drives the main gear.

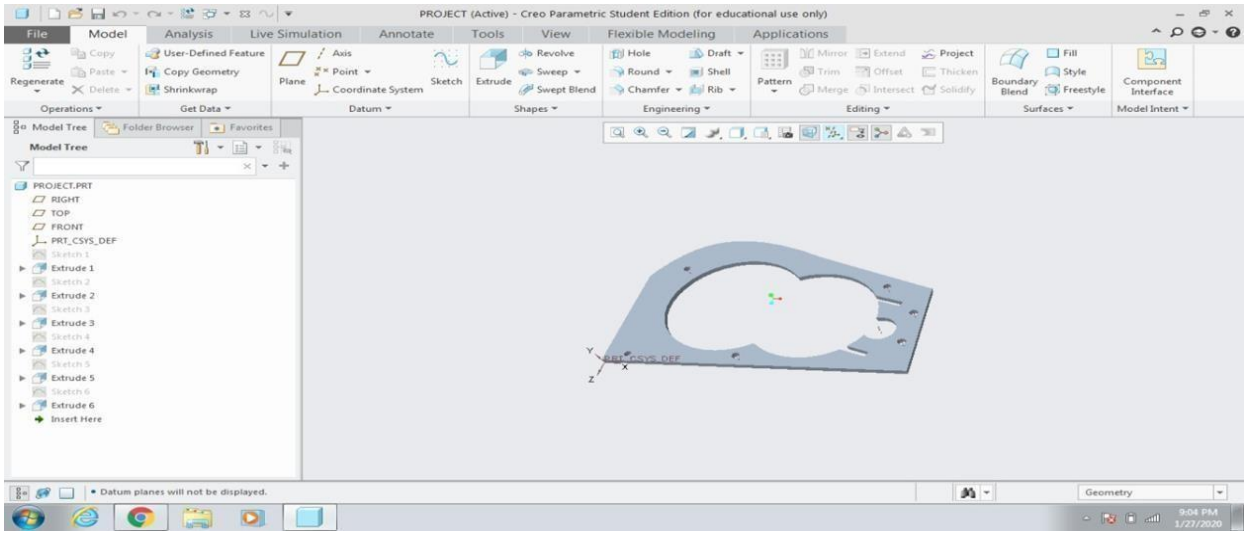


Figure.28 Design of gear box using Creo parametric software

Fabrication of gear box

- The gear box is made of thin carbon fibre sheet which is rectangular in shape. This rectangular sheet is cut into A required shape using laser cutting operation such that all the gears are mounted properly using screws and Joints and bearing on either side of the gear box.

- A gliding stopper is fixed to both side of the driven gear which is connected to a lever. This gliding stopper is used for dapping torque produced by the wing during gliding of the ornithopter.



Figure.29 Left side view of gear box

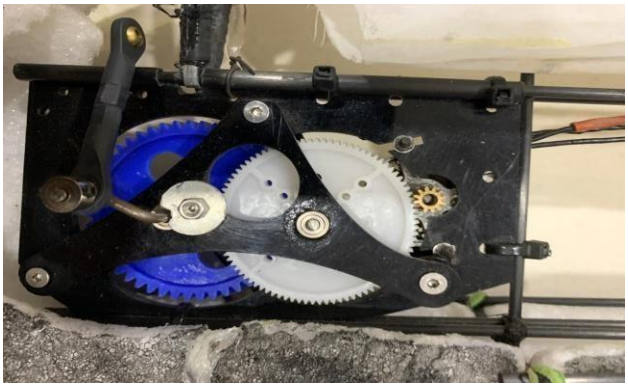


Figure.30 Right side view of gear box

- Total length of gear box = 118 mm
- Total width of gear box = 65 mm

5.8 Design of fuselage

- The fuselage, or body of the air vehicle, is a long hollow tube which holds all the pieces of an air vehicle together. The fuselage is hollow to reduce weight. As with most other parts of the airplane, the shape of the fuselage is normally determined by the mission of the aircraft. A supersonic fighter plane has a very slender, streamlined fuselage to reduce the drag associated with high speed flight.
- This fuselage holds all the mechanical and electrical parts and mechanical of the ornithopter. It is divided in to three section i.e. 1)Front section (nose) 2) Middle section (body) 3) Back end

Front section (nose)

In front section mechanical device is are allocated such as gears, gear box, BLDC motor, Barings, connecting Piston rods. The front section of the ornithopter is constructed aerodynamically to reduce drag produced by the trusses of the fuselage.

Middle section (boby)

Middle section of the fuselage contains all electrical and electronic devices such as battery, electronic speed Controller (ESC), receiver. These devices are placed at certain place places to balance the total of the fuselage.

Back end section

In back end section of the ornithopter tail is attached using threading operation and super glue. The major Weight of the fuselage is acted in back end section during flight of the ornithopter.

The material used for the fuselage is 3mm diameter and 4mm hollow diameter rods.

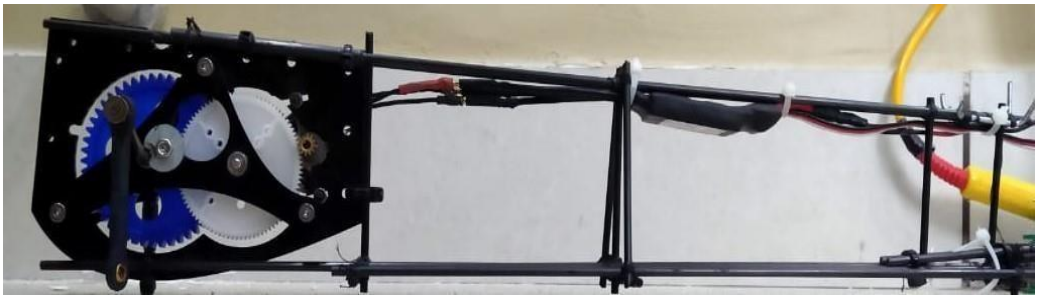


Figure.31 Fuselage of the ornithopter

Total length of the fuselage :38.5 cm

Total width of the fuselage: 6.8 cm

5.9 DESIGN OF TAIL

TAIL

A tail plane, also known as a horizontal stabiliser, is a small lifting surface located on the tail (empennage) behind the main lifting surfaces of a fixed-wing aircraft as well as other non-fixed-wing aircraft such as helicopters, gyroplanes, ornithopters.

The tail section of the ornithopter is responsible for both of the controllable degrees of freedom aside from the ability to throttle the drive motor.

The tail of the ornithopter is set up with one servo directly connected to the tail at an angle and another further up the body which rocks it via a linkage. Mounting the rudder servo at an angle is important because with a single control surfaces the elevator and rudder are naturally coupled, moving the rudder servo makes the tail also move in the vertical direction unless it's at zero angle where it doesn't have any control authority anyway. With the tail at an angle to the rudder servo it allows the servo to held into the zero-angle position with respect to the ornithopter body and while the tail stays near the trimmed position for horizontal flight.

It does not solve the problem when the elevator servo moves the tail out of the trimmed horizontal position, but being able to exploit that is a big advantage where it applies. A second point to make about the tail is that the

linkage for the elevator servo appears to be unnecessary but serves two important purposes. First is that the servo range of motion is much larger than what is necessary for the tail. By using a linkage not only does the range of motion get reduced but the mechanical advantage allows a smaller servo to be used. In addition to this, the servo, a major point of weight on the ornithopter, is moved forward in the frame. While not obviously an issue it becomes important once the centre of gravity must be placed because the frame ends up rear-heavy.

Configuration of tail

Horizontal tail and vertical tail (i.e. tails) along with wing are referred to as lifting surfaces. This name differentiates tails and wing from control surfaces namely aileron, elevator, and rudder. Due to this name, several design parameters associated with tails and wing; such as air foil, planform area, and angle of attack; are similar. Thus, several tails parameters are discussed in brief. The major difference between wing design and tail design originates from the primary function of tail that is different from wing. Primary function of the wing to generate maximum amount of lift, while tail is supposed to use a fraction of its ability to generate lift. If at any instance of a flight mission, tail nears its maximum angle of attack (i.e. tail stall angle); it indicates that there was a mistake in the tail design process. In some texts and references, tail is referred to as empennage.

The ail in a conventional aircraft has often two components of horizontal tail and vertical tail and carries two primary functions:

1. Trim (longitudinal and directional)
2. Stability (longitudinal and directional)

Since two conventional control surfaces (i.e. elevator and rudder) are indeed parts of the tails to implement control, it is proper to add the following item as the third function of tails:

3. Control (longitudinal and directional)

These three functions are described in brief here; however, more details are presented in later sections.

The first and primary function of horizontal tail is longitudinal trim; also referred to as equilibrium or balance. But the first and primary function of vertical tail is directional stability.

V tail configuration

The V-tail or Vee-tail (sometimes called a butterfly tail or Rudlicki's V-tail) of an aircraft is an unconventional arrangement of the tail control surfaces that replaces the traditional fin and horizontal surfaces with two surfaces set in a V-shaped configuration. The aft edge of each twin surface is a hinged control surface, sometimes called a ruddervator, which combines the functions of both a rudder and elevators.

Construction material for V-tail

1) Structure: carbon fiber rods (4mm)
2) Membrane: EEP (Expanded poly propylene)

Design of V- tail



Figure.32 Design of V-tail

Construction of V- tail



FIGURE. 33 Construction of V-tail

6.6 Assembly of Ornithopter

- The final assembly made by thorough fixing and binding of body parts mechanical and electrical parts.

  Fasteners, screws, stoppers and other binding material are used to assemble the body part together.

- Well inspection is carried out whether there is any loose connection or faulty assembled parts.

  A solid color is painted through out the body. A initial test is carried before actual flight of the ornithopter

- The outer frame protects from imparts on the inner parts of the ornithopter when there is any failure occurred during flight of the ornithopter.

- Further balancing is carried out by testing actual flight by observing flight path and loads acting on the ornithopter.

CHAPTER-7

FLIGHT MECHANISM OF ORNITHOPTER

Ornithopters cannot take-off from the ground directly, thus support is essential for this purpose. To fly an Ornithopter, its motion is triggered by providing it support through our hands. For initiating the motion, the power supply for motor and servos is switched on and then the whole dynamics of the ornithopter is controlled by simulator. Directions or throttle is controlled appropriately for smooth and stable flight. Rudders kept in different ways to change the direction can be explained as follows: -

1. If both the rudders are in upward direction then the Ornithopter deflects in downward direction.
2. If both the rudders are in downward direction then the Ornithopter deflects in upward direction.
3. If rightward rudder is in upward direction and leftward rudder is in downward direction then the ornithopter deflects in leftward direction.
4. If rightward rudder is in downward direction and leftward rudder is in upward direction then the ornithopter deflects in rightward direction

7.1 Ornithopter Stability and Control

Typically, the ornithopter will have a horizontal stabilizer in the back. The tail usually provides a downforce to keep the nose up, and therefore the tail incidence or angle relative to the wing is much more than you would find in an airplane. This angle is typically about 15 degrees, but it may be less or more depending on where the center of gravity is located and other aspects of the ornithopter design. Slight differences between the two wings can cause the ornithopter to turn left or right. If the wing spars are made of wood, their weights will naturally differ. You can also have asymmetry in the mechanical properties of carbon rods, or in the tension of the wing membrane. If the ornithopter wants to turn to one side, this can be corrected by adding weight to the wingtip on the outside of the turn. Through inertia, the added weight acts to suppress the flapping motion of the weighted wing, while increasing the flapping amplitude of the unweighted wing.

CHAPTER – 8

RESULTS

Table 1. Weight of each Component and Sub-assembly.

Component and Sub-assembly	Weight (g)
Lithium–Polymer Battery	30
Brushless Motor	20
Receiver	15 Electronic
Speed Controller	
2 x Servo Motor	18
2× Driving Arm and Linkage	8
Tail Structure	12
Tail membrane	7
Gear box	20
Fuselage	80
Fasteners, Screws, Nuts and Wheel-Locks	5
Wings	85
Barings	10
Outer form frame	25
Total Weight	370

1) The total weight of the ornithopter is at ideal condition is the sum of weights of all components i.e. 370 grams. This is the amount of weight should be created by ornithopter to generate lift force should be greater than the lift force **L > W.** By using flapping frequency of the wings that is **6.02 Hz**, we can achieve the required lift force and thrust to the ornithopter.

2) The velocity or the forward thrust of the ornithopter fight is controlled using speed controller which is connected to the BLDC motor which provide output power of **6.7 Watts**. Maximum velocity of the ornithopter is **20 km/h**

3) The total output power delivered by is battery is **7.5v** and **0.900 amps** that is **405 watts** per hour and input power of the electrical devices such as speed control, receiver, 2x servo motors is approximately **3500 watts** per hour.

4) So, the total time of flight of the ornithopter is given by

   = **Total power consumed by the electronics / total power delivered by the battery**

   = 3500/405

   = 8.6 mins

5) Total time of flight for the ornithopter is **7 – 9 mins**. The time of fight varies between 2 to 3 mins because of environmental factors such as wind, temperature and humidity which impacts the electrical and mechanical devices of the ornithopter.

6) The range of the flight of the ornithopter depends on the receiver and transmitter. The specification of the **flysky f6 i6** transmitter ranges between **500** meters to **800** meters.

CHAPTER-9

CONCLUSION

- In this project paper the fully functional and operated ornithopter is designed and fabricated using mechanical and electrical components such as BLDC motor, electronic speed controller, transmitter and receiver,2s battery, 2* servo motors, Gears and Gear box.
- The Ornithopter flies at a height of about 20 meters within a radius of 1 Km. from the remote system mimicking a bird as seen from ground.
- The future edition of GPs or other more exotic sensors can be possible
- Inexpensive to fabricate and assemble
- Repairs can be made quickly in the field without any need for specialized kit.
- Further research can be carried out to develop this at a large scale and increasing the thrust and overall performance

Future scope of the project

- This project can also further develop by increasing in overall flight of the ornithopter with increase in battery capacity and reduce in drag produced by the body of the ornithopter.
- Research and analysis also to be down on this ornithopter for measuring of air flow through the wings surface Which effects the thrust developed by the wings.

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